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Institute of Technology

**DEPARTMENT OF COMPUTER SCIENCE AND
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MINI PROJECT

Submitted to
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CHAPTER 1

PROJECT MANAGEMENT PLAN

1.1 GANTT CHART

A Gantt chart is a type of bar chart that illustrates a project schedule, showing the tasks, dependencies, and timelines involved. It's a popular project management tool used to plan, coordinate, and track progress.

The overall project timeline, including major tasks and deadlines, is illustrated in the Gantt Chart as shown in Figure 1.1

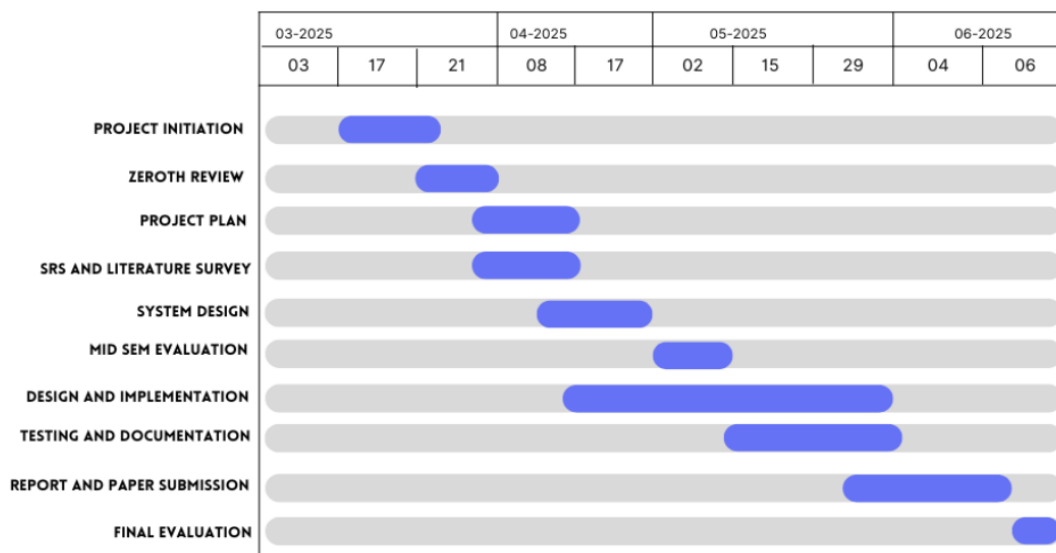


Figure 1.1 Gantt Chart showing the Project Timeline

Project Duration: March 2025 – June 2025

1. Project Initiation (March 3, 2025 – March 21, 2025)

1. Define the scope and goals of Project.
2. Identify stakeholders and form the core project team.
3. Conduct an initial briefing with the guide.
4. Outline tools, platforms, and hardware to be used.

2. Zeroth Review (March 17, 2025 - March 27, 2025)

1. Present project overview and objectives.
2. Explain Swarm Intelligence and Explainable AI relevance.
3. Gather feedback and refine the plan

3. Project Plan (March 21, 2025 - April 8, 2025)

1. Detail all work phases: planning, development, testing, submission.
2. Define timelines, responsibilities, and dependencies.
3. Finalize risk analysis and contingency strategies

4. SRS and Literature Survey (March 21, 2025 - April 8, 2025)

1. Conduct a thorough literature review on cattle health, XAI, IoT.
2. Prepare Software Requirement Specification (SRS).
3. Identify key system requirements: functional and non-functional

5. System Design (April 8, 2025 – April 17, 2025)

1. Design the architecture of the system.
2. Decide on sensors, data flow, communication protocols.
3. Create system diagrams and UI wireframes for dashboards

6. Mid-Sem Evaluation (May 2, 2025)

1. Present mid-project progress to panel.
2. Demonstrate initial setup or simulation.
3. Review and incorporate feedback into design

7. Design and Implementation (April 8, 2025 – May 29, 2025)

1. Develop data collection pipeline and cloud infrastructure.
2. Implement Explainable AI features
3. Design dashboards.

8. Testing and Documentation (May 15, 2025 – May 29, 2025)

1. Perform system testing.
2. Validate performance and system alerts.
3. Document results, testing protocols, and usage guides

9. Report and Paper Submission (May 29, 2025 - June 4, 2025)

1. Draft and finalize the project report.
2. Prepare a paper summarizing the methodology and results.
3. Submit all materials to internal and external platforms.

10. Final Evaluation (June 4, 2025 - June 6, 2025)

1. Demonstrate the final working solution.
2. Participate in evaluation, including viva and Q&A.

3. Submit source code, report, and presentation decks.

1.2 PROJECT ORGANIZATION

Project organization refers to the structure and framework used to effectively manage and successfully execute a project. project organization as shown in Table 1.2

Table 1.2 Project Organization

Process Model	Chosen	Justification
Agile Model	Yes	Best suited for dynamic projects like this one where requirements and insights evolve based on data
Scrum Framework	Yes	Works well with Agile to manage tasks using sprints, stand-ups, and continuous feedback.
Iterative Model	Yes	Each version of the ML framework improves upon the previous based on results and evaluation.
Waterfall Model	Partially	Used only for fixed, early-stage phases like requirement gathering and hardware/software planning

1.3 ROLES AND RESPONSIBILITIES

Roles and responsibilities refer to the specific duties, tasks, and expectations assigned to individuals or teams within an organization or project.

The roles and responsibilities assigned to each team member are detailed in Table 1.3

Table 1.3 Roles and Responsibilities

Name	Role	Responsibilities
HARIPRASAD B R	Data Engineer	Data collection, cleaning, storage setup

ARYAN KUMAR	Cognitive Analyst	Integrating psychology, theories, interpreting bias patterns
I SUPRABATH REDDY	ML Engineer	Model selection, Training, Tuning
BASAVARAJ BHAJANTRI	Visualization	Building network graphs, visualizing sentiment analysis

1.4 WORK BREAKDOWN STRUCTURE

A Work Breakdown Structure (WBS) is a hierarchical decomposition of the total scope of work to be carried out by the project team to accomplish the project. The overall process followed in the project is visually represented through the Work Breakdown Structure (WBS) in Figure 1.4.

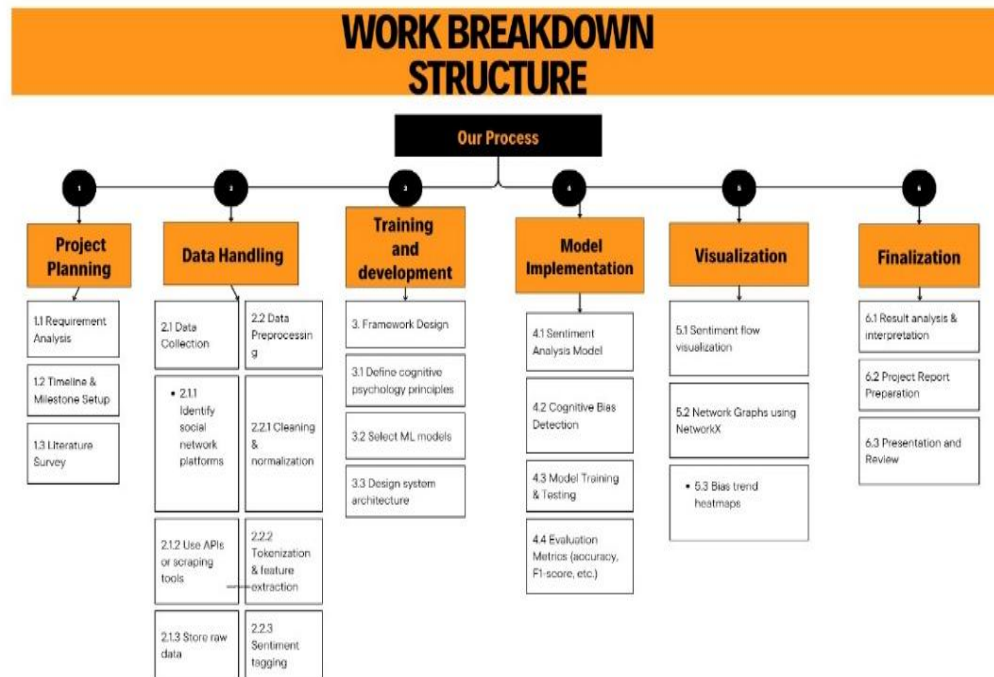


Figure 1.4 Work Breakdown Structure

1. Project Planning

This phase involves setting the foundation for the project:

- 1.1 Requirement Analysis: Understanding what the project needs to accomplish
- 1.2 Timeline & Milestone Setup: Defining key dates and delivery points.
- 1.3 Literature Survey: Researching existing work to understand the background and gaps.

2. Data Handling

All tasks related to acquiring and preparing data:

- 2.1 Data Collection
 - 2.1.1 Identify social network platforms: Decide platforms like Twitter, Reddit, etc.
 - 2.1.2 Use APIs or scraping tools: Collect data programmatically.
 - 2.1.3 Store raw data: Save collected data for processing
- 2.2 Data Preprocessing
 - 2.2.1 Cleaning & normalization: Remove noise, make data consistent
 - 2.2.2 Tokenization & feature extraction: Break text into tokens, extract useful features.
 - 2.2.3 Sentiment tagging: Label data with positive/negative/neutral sentiments.

3. Framework Design

Focuses on how the system will work:

- 3.1 Define cognitive psychology principles: Theories that guide analysis of human behavior or bias
- 3.2 Select ML models: Choose appropriate machine learning models.
- 3.3 Design system architecture: Outline how the components interact (data flow, APIs, models, etc.).

4. Model Implementation

Where development and testing of models happens:

- 4.1 Sentiment Analysis Model: Classify sentiment in text (positive, negative, neutral)
- 4.2 Cognitive Bias Detection: Detect biases in the language used
- 4.3 Model Training & Testing: Train models using labeled data and evaluate performance.
- 4.4 Evaluation Metrics (accuracy, F1-score, etc.): Use metrics like accuracy, F1-score to measure how well models perform.

5. Visualization

Make the results understandable through visuals

- 5.1 Sentiment flow visualization: Show how sentiment changes over time or topics.
- 5.2 Network Graphs using Network X: Visualize relationships between entities (e.g., people, topics).
- 5.3 Bias trend heatmaps: Visualize where and how often bias appears in data.

6. Finalization

Wrap-up and reporting:

- 6.1 Result analysis & interpretation: Make sense of the outcomes.
- 6.2 Project Report Preparation: Document all work done, methods used, and results.
- 6.3 Presentation and Review: Share findings with stakeholders and get feedback.

CHAPTER 2

LITERATURE SURVEY

2.1 Introduction to the Literature Survey

In today's digital landscape, technology increasingly mediates our information access, decision-making, and social interactions, yet many systems prioritize technical optimization over psychological compatibility. This literature survey explores an emerging interdisciplinary field that integrates psychological principles into computational systems to create more human-centered technology. By understanding cognitive processes, emotional responses, and social behaviors, researchers are developing systems that work in harmony with human psychology rather than exploiting or ignoring it.

The fifteen key research papers (2017-2024) reviewed here demonstrate the field's evolution from theoretical foundations in recommender systems to diverse applications addressing critical challenges like cyberbullying detection, misinformation spread, and data overload. These works collectively reveal an important paradigm shift: moving from engagement-focused metrics toward psychological well-being as a design objective, developing more sophisticated methods for implementing psychological insights in technical systems, and increasing attention to the ethical implications of psychologically-informed technology. This integration of machine learning and cognitive psychology promises a digital future where technology enhances human capabilities while respecting cognitive limitations and supporting emotional welfare.

2.2 Related Works

Related works are as shown in the below Table 2.1

Table 2.1 Related Works

S L	Paper	Methodology/ Technologies	Performance Measurement	Values	Comment/ Future
1	Lex, E., Kowald, D., Seitlinger, P., Tran, T. N. T., & Schedl, M. (2021). Psychology-informed recommender systems. Foundations and Trends® in Information Retrieval, 15(1), 1-87.	Review of psychology theories applied to recommender systems; Integration frameworks for cognitive processes	Survey metrics; Literature review coverage	Interdisciplinary approach; Human-centric design	Foundation work establishing the field of psychology-informed recommender systems
2	Shen, X., Hu, Z., Chen, Q., Liu, S., Liang, R., & Sun, J. (2024). Evolvable psychology-informed neural network for memory behavior modeling. arXiv preprint arXiv:2401.12345.	Evolvable neural networks; Cognitive memory modeling; Adaptive learning	Memory recall accuracy; Temporal cognition modeling; Adaptation rates	Biological plausibility; Cognitive fidelity	Proposes evolving architectures that mirror human memory processes for more natural AI systems
3	Lex, E., & Schedl, M. (2022). Psychology-informed recommender systems tutorial. Proceedings of the 16th ACM Conference on Recommender	Tutorial methods; Educational frameworks for psychology-RS integration	Tutorial effectiveness; Knowledge transfer metrics	Educational value; Interdisciplinary knowledge sharing	Focuses on training the next generation of researchers in this interdisciplinary field

	Systems (RecSys'22), 1-5.				
4	Schedl, M., Moscatti, M., Sguerra, B., & Lex, E. (2024). Psychology-informed information access systems workshop. Proceedings of the ACM Web Search and Data Mining Conference (WSDM'24), 1-3.	Workshop methodologies; Collaborative research frameworks	Participation metrics; Research impact assessment	Community building; Knowledge exchange	Workshop aimed at building community and establishing research priorities
5	Hall, D. L., Silva, Y. N., Wheeler, B., Cheng, L., & Smith, R. (2022). Harnessing the power of interdisciplinary research with psychology-informed cyberbullying detection models. International Journal of Data Science and Analytics, 6(4), 245-263.	Cyberbullying detection models; Social psychology frameworks; NLP techniques	Detection accuracy; False positive rates; Psychological validity	Protection of vulnerable users; Ethical AI implementation	Applies psychological insights to improve detection of harmful content while reducing false positives
6	Bhowmick, S. S., Li, H., Chen, S. H., A., & Zhao, Y. (2024). Social psychology meets social computing: State of the art and future directions. Companion	Integration of social psychology theories with computing systems; Human-centered modeling	Social interaction metrics; User behavior prediction accuracy	Social welfare; User empowerment	Maps the intersection between social psychology and computing with future research directions

	Proceedings of the ACM Web Science Conference (WebSci'24), 1-10.				
7	Bhowmick, S. S., Chen, S. H. A., & Srivastava, D. (2024). Cognitive psychology meets data management: State of the art and future directions. Companion of the 2024 ACM International Conference on Data Engineering (ICDE'24), 1-8.	Cognitive psychology principles in data management; User cognition models	Data comprehension metrics; Cognitive load measurement	Cognitive accessibility; User understanding	Explores how data management systems can be designed to align with human cognitive processes
8	Lex, E., & Schedl, M. (2022). Psychology-informed recommender systems: A human-centric perspective on recommender systems. Proceedings of the ACM Conference on Human Information Interaction and Retrieval (CHIIR'22), 1-7.	Human-centric recommender systems; Psychological user modeling	User satisfaction; Decision quality; Psychological well-being	Human autonomy; Mental well-being	Emphasizes designing systems that support rather than manipulate human decision-making
9	Henry, N. F. (2023). Modelling music selection in everyday life with applications for psychology-informed	Music selection modeling; Ecological momentary assessment;	Music preference prediction; Situational appropriateness	Personal expression; Emotional well-being	PhD thesis developing models that account for psychological

	d music recommender systems. Ph.D. Thesis, University of Leeds.	Context-aware recommendations			context in music selection
10	Bozdemir, B. S. (2022). Psychology and social policy. Springer.	Psychological frameworks for social policy; Behavioral intervention design	Policy effectiveness; Behavioral change metrics	Social welfare; Evidence-based policy	Book connecting psychological insights to effective social policy development
11	Allcott, H., & Gentzkow, M. (2017). Social media and fake news in the 2016 election. Journal of Economic Perspectives, 31(2).	Economic analysis of fake news; Survey methodology; Statistical modeling	News consumption patterns; Belief accuracy metrics	Democratic integrity; Information quality	Foundational study quantifying the impact of fake news in electoral contexts
12	Vosoughi, S., Roy, D., & Aral, S. (2018). The spread of true and false news online. Science, 359(6380), 1146-1151.	Large-scale information diffusion analysis; Social network mapping	Information spread rates; Truth vs. falsehood comparison	Truth-seeking; Information integrity	Demonstrated that false information spreads faster than true information online
13	Pennycook, G., & Rand, D. G. (2021). The psychology of fake news. Trends in Cognitive Sciences, 25(5), 388-402.	Cognitive psychology of misinformation; Analytical thinking assessment	Belief accuracy; Cognitive reflection test correlation	Critical thinking; Media literacy	Review highlighting cognitive factors in fake news susceptibility
14	Shu, K., Sliva, A., Wang, S., Tang, J., & Liu, H. (2017). Fake news detection	Data mining for fake news detection; Feature importance analysis	Detection accuracy; Feature importance analysis	Information integrity; Social	Data-centric approach to automated

	on social media: A data mining perspective. ACM SIGKDD Explorations Newsletter, 19(1), 22-36.	engineering; Social context modeling		responsibility	detection of misinformation
15	Kumar, A., & Sharma, A. (2021). A review on sentiment analysis and emotion detection from text. Journal of the Indian Institute of Science, 101(4), 435-448.	Sentiment analysis techniques; Emotion detection algorithms; NLP methods	Sentiment classification accuracy; Emotion detection precision	Emotional understanding; Communication quality	Review of text-based approaches to understanding human emotional states

2.3 Conclusion of survey

The Future of Psychology-Informed Systems: A Promising Interdisciplinary Frontier

Integrating cognitive and social psychology into computational systems marks a major shift toward more human-centered technology. From early work in psychology-based recommender systems to recent applications in cyberbullying detection, misinformation tracking, and emotional text analysis, this interdisciplinary approach is addressing both technological and societal challenges.

Advancements in neural networks that reflect human memory and cognition are pushing these systems beyond efficiency toward ethical and accessible design. With growing focus on human autonomy, emotional well-being, and information integrity, psychology-informed systems are shaping a future where technology aligns closely with human needs and values.

CHAPTER 3

SOFTWARE REQUIREMENT SPECIFICATIONS

3.1 Introduction

With the exponential rise of social media platforms, understanding how information spreads and influences human cognition has become critical. This software aims to bridge the gap between data-driven analysis and cognitive psychological theories by leveraging machine learning algorithms. By incorporating cognitive biases and behavioral tendencies into analytical models, the system aspires to provide deeper insights into how users perceive, process, and propagate information online. This document captures the technical and functional requirements necessary to realize such a multidisciplinary system.

3.1.1 Purpose

The purpose of this document is to define the software requirements for the system titled *“A Cognitive Psychology Approach to Analyze Information in Social Network Using Machine Learning.”* This SRS serves as a blueprint that clearly outlines what the software is intended to do, the constraints under which it must operate, and the features it should include.

The system is designed to analyze information dissemination and cognitive behavior on social networks by using machine learning (ML) techniques informed by cognitive psychology. The ultimate goal is to detect behavioral patterns, cognitive biases. This document will guide the development team in implementing the desired functionality and will serve as a baseline for validation, verification, and testing.

Intended Audience:

- **Software Developers** – To build and implement the system as per defined requirements.
- **Researchers and Psychologists** – To understand how cognitive psychology models are integrated into ML-based social media analysis.
- **Project Supervisors and Evaluators** – To assess whether the delivered system meets its intended objectives.
- **Testers** – To create test cases and validate functionality and performance.

3.1.2 Scope

The proposed system is a tool that will analyze user-generated content using Natural Language Processing (NLP) and classification models to extract psychological insights. The software will perform tasks such as sentiment analysis, bias detection, misinformation recognition, and user profiling based on cognitive psychology principles.

The software will:

- Analyze content to detect cognitive distortions, sentiment shifts, and misinformation patterns.
- Classify users dynamically using psychology-inspired machine learning models.
- Present insights in graphical and tabular formats.
- Provide interactive capabilities through a memory-augmented chatbot.

3.1.3 Product scope

The proposed software is a cognitive analytics platform that leverages Natural Language Processing (NLP) and machine learning to derive psychological insights from social network content. It is designed to detect cognitive distortions, emotional shifts, biases, and misinformation in user-generated data, enabling a deeper understanding of online behavior through a cognitive psychology lens.

The system provides dynamic user profiling based on both stable personality traits and evolving emotional states. Key features include an AI-powered chatbot with long-term memory for personalized interaction, as well as visual dashboards for insight presentation. This tool serves as a valuable asset for psychologists, cognitive scientists, and data analysts exploring behavior in digital spaces.

By aligning with goals such as mental health support, ethical AI development, and intelligent content analysis, the software supports research, education, and strategic business insights. It also promotes responsible AI use in understanding and responding to human behavior on social platforms.

This system is particularly beneficial for:

- **Psychologists and Cognitive Scientists** – for behavioral analysis.
- **Data Scientists** – for experimenting with psychologically inspired ML models.

3.2 General Description

This section outlines the general characteristics of the software product, including its context, functionality, and user base. It describes what the system is intended to do from the user's point of view, highlights its importance, and provides a broad understanding of the expected operating conditions, constraints, and dependencies.

3.2.1 Product Perspective

The proposed system, “A Cognitive Psychology Approach to Analyse Information in Social Network Using Machine Learning,” is a new, self-contained product that does not

rely on or extend any existing commercial software systems. It is designed as a standalone analytical tool that integrates machine learning algorithms with cognitive psychology principles to process and interpret social media data.

The system's components are modular in nature, comprising:

- Data Collection Module
- Preprocessing & NLP Module
- ML-Based Analysis Engine
- Cognitive Profiling System
- Interactive Memory-Augmented Chatbot
- Visualization Dashboard

These components interact in a pipeline-based architecture, where each module's output serves as input for the next. This modular approach supports reusability and independent updates. The system interfaces only with publicly available APIs and does not require integration with existing enterprise or governmental systems.

3.2.2 Product Functions

The software will provide the following high-level functions:

- **Data Scraping and Ingestion:** Filters based on language, hashtags, timeframes, or user handles.
- **Natural Language Processing and Preprocessing:** Performs tokenization, lemmatization, noise removal, and language detection. Identifies sentence structures and emotional tones.
- **Sentiment and Bias Analysis:** Detects sentiments (positive, negative, neutral). Identifies psychological biases like confirmation bias, framing, anchoring, etc.
- **User Profiling Based on Cognitive Patterns:** Classifies users into psychological profiles based on their language use and behavioral patterns. Updates profiles over time using reinforcement learning.
- **Misinformation and Distortion Recognition:** Uses classifier models to detect potentially misleading or emotionally manipulative content.
- **Interactive Visualization:** Displays data as graphs, pie charts, and trend lines. Offers downloadable reports and CSV summaries.
- **Memory-Augmented Chatbot:** Engages with users through a conversational UI. Retains past conversations to improve recommendations and explanations.

3.2.3 User Classes and Characteristics

The system is designed for multiple user groups, each with different expertise and usage patterns as in Table 3.1:

Table: 3.1 User classes

User Class	Characteristics
Software Developers	Technical users who will implement or maintain the system. Familiar with APIs, ML.
Psychologists / Researchers	Moderate technical expertise. Focused on interpreting behavioral data and biases.
Data Scientists	Proficient in ML, interested in model performance, datasets, and statistical output.
Supervisors / Evaluators	High-level users assessing project progress and outcomes. Limited technical depth.
End Users (Chatbot Users)	General users interacting with the chatbot for insights. Require high usability.

Priority User Classes:

- Researchers and psychologists (core insights consumers)
- Data scientists (model fine-tuning and experimentation)

3.2.4 Operating Environment

The system is intended to run in the following environment:

- **Hardware:** Standard desktop/laptop with minimum 8 GB RAM, 2 GHz processor, or hosted on cloud VMs (e.g., AWS EC2).
- **Operating Systems:** Windows 10/11, Ubuntu 20.04 or later
- **Programming Language:** Python 3.8+
- **Software Dependencies:** Scikit-learn, TensorFlow, NLTK, SpaCy, Hugging Face Transformers, Flask or Streamlit for UI/dashboard
- **Cloud Compatibility:** AWS/GCP for deployment and storage
- **Database:** SQLite or PostgreSQL for chatbot memory and user logs

3.2.5 Design and Implementation Constraints

The following constraints will guide and limit development:

- **Language & Framework Constraints:** All back-end components must be developed in Python. Only open-source libraries are preferred due to budget constraints.
- **Hardware Constraints:** Must operate efficiently on mid-range systems or low-cost cloud platforms.
- **Ethical and Regulatory Constraints:** Must comply with data privacy norms (e.g., avoid personal identifiers). Follows ethical AI practices: transparent model reasoning, bias mitigation, and explainability.
- **Development Constraints:** The entire project must be completed within 10 weeks.

3.2.6 User Documentation

The following documentation components will be provided:

- **User Manual (PDF):** Explains installation, configuration, usage of each module, chatbot interaction, and troubleshooting.
- **Online Help / Tooltips:** Integrated in UI, explaining graphs, user profiles, and model outcomes.
- **API Usage Guide:** For advanced users wanting to reuse or extend functionality.
- **Tutorial Videos:** Short walkthroughs for setting up the system and interpreting outputs

3.2.7 Assumptions and Dependencies

The system development is based on the following assumptions:

Assumptions:

- The dataset collected from public social media is sufficient for meaningful insights.
- Users will have a basic understanding of English for interacting with the chatbot and dashboard or cross linguistics is also provided.
- No major changes occur in library support (e.g., deprecation of NLP packages).

Dependencies:

- MDTI for data collection.
- Python libraries for machine learning and NLP (e.g., scikit-learn, TensorFlow, Hugging Face).

- Availability of cloud platforms or virtual environments for model training and deployment.

3.3 Functional Requirements

3.3.1 User Class A: Analyst/Researcher

Analysts or researchers interact with the system to collect, preprocess, and analyze text data from the MBTI dataset. They leverage NLP and machine learning tools to detect biases, sentiments, and build psychological user profiles for behavioral studies.

- **Feature A – Data Collection**
Allows users to collect and store text-based data from the MBTI dataset for studying online behavior.
- **Feature B – Data Preprocessing**
Prepares raw social media data for analysis by cleaning, normalizing, and structuring it for NLP models.
 - **Stimulus/Response Sequence:**
 - *Stimulus:* User selects a dataset and clicks "Preprocess Data."
 - *Response:* System removes stop words, emojis, URLs, and performs tokenization.
 - **Associated Functional Requirements:**
 - **FR-B1:** The system shall clean the text by removing non-textual elements.
 - **FR-B2:** The system shall normalize text (e.g., lowercasing, stemming).
 - **FR-B3:** The system shall validate the output format for compatibility with the NLP pipeline.
- **Feature C – NLP & Bias Detection**
Detects sentiment and cognitive biases such as confirmation bias, framing, or emotional appeals in posts.
 - **Stimulus/Response Sequence:**
 - *Stimulus:* User submits cleaned text for analysis.
 - *Response:* The system runs NLP and ML models to output sentiment scores and bias classifications.
 - **Associated Functional Requirements:**
 - **FR-C1:** The system shall extract sentiment polarity and subjectivity.

- **FR-C2:** The system shall identify and label cognitive distortions.
 - **FR-C3:** The system shall generate an analysis report.
- **Feature D – User Profiling via ML**

Creates psychological user profiles using linguistic features and behavioral patterns.

 - **Stimulus/Response Sequence:**
 - *Stimulus:* NLP analysis is completed.
 - *Response:* The system classifies users into psychological archetypes using supervised or reinforcement learning.
 - **Associated Functional Requirements:**
 - **FR-D1:** The system shall use trained ML models to assign users to cognitive profiles.
 - **FR-D2:** The system shall adapt user profiles over time based on new inputs.
 - **FR-D3:** The system shall store user profiles securely and link them to analysis sessions.

3.3.2 User Class B: End-User (Chatbot Interactor)

End-users interact with a memory-augmented chatbot to receive personalized, context-aware responses and behavior-based insights.

- **Feature A – Memory-Augmented Chatbot**

Provides an intelligent conversational interface that uses historical context for meaningful interaction.

 - **Stimulus/Response Sequence:**
 - *Stimulus:* User sends a query via chatbot.
 - *Response:* System retrieves context from memory and responds intelligently.
 - **Associated Functional Requirements:**
 - **FR-A1:** The chatbot shall maintain conversation history per user.
 - **FR-A2:** The chatbot shall provide personalized responses based on past interactions.
 - **FR-A3:** The chatbot shall allow users to request behavior-based predictions.
 - **FR-A4:** The system shall handle unexpected inputs gracefully and log errors.

3.3.3 User Class C: Administrator

Administrators manage and monitor the system's performance, configuration, and user activity via a backend dashboard.

- **Feature A – System Monitoring and Configuration**

Enables admins to oversee system health, logs, and configurations.

- **Stimulus/Response Sequence:**

- *Stimulus:* Admin logs into backend dashboard.
- *Response:* System displays logs, analytics, and configuration options.

- **Associated Functional Requirements:**

- **FR-A1:** The system shall log all system events and errors.
- **FR-A2:** The system shall provide performance metrics and user activity stats.
- **FR-A3:** The system shall allow admin to configure API keys, data sources, and model parameters.

3.4. Interface Requirements

This section outlines how the software interacts with users, hardware, other software components, and communication protocols. It describes all logical and physical interfaces, ensuring smooth and predictable operations of the system within its environment.

3.4.1 User Interfaces

The system will feature a clean, user-friendly Graphical User Interface (GUI) with the following characteristics:

- **Home Dashboard:** Displays a summary of analytics (e.g., sentiment charts, bias heatmaps, misinformation alerts). Allows users to initiate new analyses or view historical results. Contains a top navigation bar with tabs: *Dashboard*, *Live Analysis*, *Chatbot*, *Settings*, *Help*.
- **Data Analysis Panel:** Offers input boxes or upload features to add data sources (e.g., keywords, hashtags, or CSV files). Displays progress of data scraping and analysis in real time via status bars or progress indicators.
- **Chatbot Interface:** Memory-augmented chatbot interface with a chat window and typing bar. Remembers context and offers real-time psychological insights based on user inputs.
- **Standard UI Elements**
 - **Buttons:** *Submit*, *Reset*, *Download Report*, *Back*, *Next*.
 - **Keyboard Shortcuts:** (e.g., Ctrl+Enter for submit, Esc to exit screens).

- **Tooltips and Help Popups:** Present across all input fields for clarity.
- **Error Messages:** Will follow consistent format in pop-up or inline form (e.g., "Invalid input: Please enter a valid keyword.").
- **Design Standards:**Color scheme: Professional palette (e.g., blue, white, grey).Font: Roboto or Arial, 14pt for normal text, 18pt bold for headings.GUI design will follow standard UI/UX principles and accessibility guidelines (WAI-ARIA).

A separate *User Interface Specification* document will include sample screen designs and layout sketches.

3.4.2 Hardware Interfaces

The system is designed to run on general-purpose computing hardware. The following hardware interface requirements are defined:

- **Client Devices:**
 - Standard desktop/laptop systems.
 - Recommended system configuration:
 - CPU: Minimum 2.5 GHz dual-core processor.
 - RAM: 8 GB minimum.
 - Storage: 5 GB free space.
- **Optional GPU Support:**If running advanced NLP tasks locally, optional GPU acceleration (e.g., NVIDIA CUDA-enabled GPU) is supported for models like BERT.
- **Input Devices:**Mouse and keyboard.Optional: Microphone (for potential voice-based chatbot integration in future releases).
- **Output Devices:**Standard displays with minimum 1366×768 resolution.PDF and CSV export options supported via connected printers or file systems.

3.4.3 Software Interfaces

The software interacts with various external software systems, libraries, and APIs. These interfaces include:

- **Operating System:**Platform-independent; runs on Windows, macOS, or Linux.Recommended: Ubuntu 20.04+ / Windows 10+.
- **Programming Languages and Frameworks:**Python 3.9+ (primary backend and ML scripting language).Flask/Django (for web-based interface). JavaScript with React (for dynamic frontend components).
- **Libraries and Tools:**
 - **NLP/ML:**scikit-learn (v1.3+), TensorFlow (v2.10+), HuggingFace Transformers.

- **Visualization:**Matplotlib, Plotly, Seaborn.
- **Data Handling:**pandas, NumPy.
- **Chatbot Memory System:**SQLite or TinyDB for session memory storage.
- **Databases:**MongoDB (preferred for storing structured insights).
- **Inter-process Communication:**JSON-based messaging format.RESTful endpoints to support modular architecture.

3.4.4 Communications Interfaces

The system communicates with external services and clients through secure and standardized communication protocols.

- **Client-Server Communication:**RESTful APIs with HTTP/HTTPS protocols.All requests/responses in JSON format.
- **Security Measures:**Token-based authentication for API access.HTTPS for encrypted transmission.Rate limiting and API key management to prevent abuse.
- **Error Handling and Synchronization:**Status codes: 200 (OK), 400 (Bad Request), 404 (Not Found), 500 (Internal Server Error).Retry mechanisms for temporary outages or network delays.Sync mechanisms to update client dashboards with latest data in near real-time.

3.5 Performance Requirements

This section outlines the measurable performance expectations of the software system under specified operating conditions. It includes both **static numerical requirements** (such as the number of users and data capacity) and **dynamic numerical requirements** (such as response time and processing speed). All values are either based on testing projections or industry standards and are stated in precise, quantifiable terms.

3.5.1 Static Numerical Requirements

These requirements pertain to fixed system capabilities regardless of workload fluctuations.

a) Number of Terminals Supported

- The system will support up to **50 concurrent client terminals** (browser-based sessions) connected to the server without degradation in performance.
- These terminals may access functionalities like live data analysis, chatbot interactions, and report viewing simultaneously.

b) Simultaneous Users

- The system shall support at least **30 simultaneous authenticated users** performing different operations under average network conditions.
- The architecture will be scalable using containerized deployments (e.g., Docker, Kubernetes) to accommodate future user growth up to 100 concurrent users.

3.5.2 Dynamic Numerical Requirements

These requirements describe how the system behaves under typical and peak conditions.

a) Response Time

- 95% of standard user requests (e.g., dashboard load, analysis report retrieval) shall complete in **under 2 seconds**.
- The live analysis feature shall begin displaying real-time results within **5 seconds** of data stream initiation.

b) Data Processing Time

- **Preprocessing (cleaning and tokenization)** of up to 10,000 posts shall complete in **less than 30 seconds**.
- **Machine learning-based inference** (sentiment, bias, psychological trait prediction):
 - Shall process 1,000 posts in **under 20 seconds** on standard hardware (8 GB RAM, 4-core CPU).
 - GPU-accelerated systems shall reduce this to **under 5 seconds** for the same load.

c) System Throughput :The chatbot module can handle **10 conversations per second**, with low-latency access to memory (within 0.5 seconds per memory retrieval).

d) Error Rate

- System accuracy for ML models (based on benchmark datasets):
 - Sentiment Analysis: **≥ 90% accuracy**.
 - Bias Detection: **≥ 85% precision and recall**.
 - Cognitive Trait Classification: **≥ 80% F1-score**.
- Maximum allowable system failure rate (unhandled exceptions, crashes) during normal operation: **< 0.5% of sessions**

3.5.3 Capacity Planning and Scalability

- The backend will be cloud-ready, and horizontal scaling is supported to accommodate spikes (e.g., trending topic analysis).

- Cloud load balancers and caching mechanisms will ensure consistent response times during peak loads.
- Capacity targets:
 - Short-term: 50,000 posts/day with 30 users.
 - Long-term (1 year): Scalable up to 500,000 posts/day and 100 users.

3.5.4 Performance Testing Conditions

To validate the above, performance testing will be conducted under the following scenarios as in Table 3.2:

Table 3.2 Performance

Scenario	Simultaneous Users	Data Volume	Expected Time
Light Load	10 users	1,000 posts	< 5 seconds
Normal Operation	30 users	10,000 posts	< 30 seconds
Peak Load (stress testing)	100 users	100,000 posts	< 3 minutes

3.6 Design Constraints

This section outlines the limitations and restrictions imposed on the design and implementation of the system. These constraints may originate from hardware, software, data handling regulations, chosen algorithms, or compliance with specific standards. They define boundaries within which the system must be developed, ensuring compatibility, maintainability, and regulatory adherence.

3.6.1 Hardware Constraints : The following are the hardware constraints

- **Minimum Server Specifications:**CPU: 4-core (Intel i5/Ryzen 5 equivalent or higher). RAM: 8 GB (16 GB recommended for high-load analysis). Storage: SSD with minimum 256 GB for fast read/write operations. GPU: Optional, but required for deep learning-based inference (NVIDIA CUDA-enabled with 4GB VRAM minimum)
- **Client Device Compatibility:**The system must be fully usable on modern browsers (Chrome, Firefox, Edge) on desktops and mobile devices. No installation should be required beyond the browser.

3.6.2 Software Constraints : The following are the software constraints

- **Programming Languages:**
 - Backend: Python 3.9+ (mandatory for ML libraries and data processing)

- Frontend: HTML5, CSS3, JavaScript (React.js or equivalent)
- Database: SQLite for local development, PostgreSQL/MySQL for deployment
- **Frameworks & Libraries:**
 - TensorFlow / PyTorch for cognitive ML models
 - Scikit-learn for preprocessing and feature engineering
 - Flask / FastAPI for backend API endpoints
 - NLTK / SpaCy for NLP tasks
- **Operating System Compatibility:**
 - Development and deployment will be Linux-based (Ubuntu 20.04+), but must support containerization for platform-independence using Docker.

3.6.3 Algorithmic and Model Constraints: The following are the algorithmic and model constrain

- **Algorithm Choice:**
 - Models must support explainability using SHAP or LIME, as interpretability is essential for psychological analysis.
 - Sentiment and cognitive trait analysis must use **lightweight ML models** (e.g., logistic regression, decision trees, or compact neural networks) unless GPU is available.
 - Online learning or batch inference required—no continuous learning mode will be implemented to ensure consistency.
- **Data Privacy Limitation:**
 - Data must be anonymized and comply with ethical research guidelines.
 - Personally identifiable information (PII) cannot be stored or inferred from analysis.

3.6.4 Integration Constraints: The following are the Integration constraints

- **Database Schema:**
 - The data model must support efficient storage of text, metadata, and analysis outputs (e.g., timestamps, sentiment scores, psychological labels).
 - All data must be timestamped and linked to the original post/user ID (anonymized).

3.6.5 Standards Compliance

This subsection details the standards and procedures that the software must adhere to for uniformity, legality, and traceability.

a) Report Format: All system-generated reports must

- Be in **PDF and CSV formats**
- Include **timestamp, summary statistics, model confidence scores**, and visualizations (charts, word clouds)
- Follow a predefined structure with **title, methodology, results, interpretation, and conclusion sections**

b) Data Naming Conventions

- Variables and fields must follow `snake_case` format (e.g., `sentiment_score`, `cognitive_label`) for consistency.
- Dataset filenames must include date, source platform, and topic (e.g., `2025-04-07_twitter_climatechange.csv`).

c) Accounting and Logging Procedures

- The system must maintain **comprehensive logs** for all backend processing operations.
- Logs must capture:
 - API calls made (timestamp, endpoint, user)
 - ML model inference (model used, input sample size, time taken)
 - Errors and exceptions with tracebacks

d) Audit Tracing

- All user actions (e.g., data upload, analysis initiated, model configuration changes) must be recorded.
- Changes to any data (e.g., deletion, reclassification) must be auditable, including:
 - **Before and after values**
 - **User identity** (anonymized if necessary)
 - **Timestamp and action type**

3.7 Non-Functional Attributes

This section outlines the critical non-functional requirements that the system must fulfill to ensure effective performance, security, usability, maintainability, and compliance. While functional requirements describe *what* the system should do, non-functional attributes describe *how well* the system should perform under various constraints and conditions.

3.7.1 Performance Requirements

These requirements define how the system must perform in terms of speed, resource usage, and responsiveness across various operating conditions.

- **Response Time:** 95% of sentiment and cognitive label inferences must be completed within **1 second** for datasets with fewer than 1000 entries. For larger datasets (1000–10,000 entries), batch processing must complete within **30 seconds** on standard server hardware.
- **Concurrent Users:** The system must support at least **50 concurrent users** without performance degradation. For each user, memory consumption must not exceed **200MB** during peak usage.
- **Throughput:** The system should handle at least **10,000 records** per hour during peak load scenarios.
- **Resource Utilization:** CPU usage should not exceed 80% under high load. GPU acceleration (if used) should be leveraged for deep learning-based inference without overloading the server.
- **Rationale:** Performance is critical for real-time analysis tasks in psychology-focused social media research. Rapid inference and feedback enhance the usability and responsiveness for academic or analyst users.

3.7.2 Safety Requirements

These requirements address any potential for harm, misuse, or critical data loss.

- **Data Safety:** Regular automatic backups must be implemented at 12-hour intervals to prevent data loss. All user-uploaded data must be **validated and sanitized** before processing to avoid system compromise.
- **Access Restrictions:** Only authorized users should be able to perform administrative actions like data deletion, model updates, or system reconfiguration.
- **Error Handling:** The system must fail gracefully with informative error messages and log events when encountering malformed inputs, missing data, or API failures.
- **Compliance:** The system must comply with data handling standards relevant to academic research involving human data (e.g., IEEE ethics, institutional IRB standards).

3.7.3 Security Requirements

Security ensures the protection of data, models, and infrastructure from unauthorized access, manipulation, or breaches.

- **Authentication:** Multi-level authentication must be implemented (e.g., role-based login for Admin, Analyst, and Viewer). Passwords must be stored using strong encryption algorithms (e.g., bcrypt or SHA-256 with salting).
- **Data Privacy:** Personally Identifiable Information (PII) must never be stored. All data must be **anonymized** and handled in accordance with GDPR-like data protection principles.

- **Data Transmission:**All data exchanges between client and server must be secured using **HTTPS** with valid SSL certificates.
- **Vulnerability Protection:**The system must be protected against common security threats (e.g., SQL injection, XSS, CSRF) using input sanitization and secure coding practices.
- **Security Auditing:**Logs of all access and administrative operations must be maintained for auditing.

3.7.4 Software Quality Attributes

These describe the overarching quality factors that enhance the user and developer experience:

- **Reliability:**The system must maintain 99% uptime with automatic recovery from crashes.A watchdog process should be used to restart crashed services if necessary.
- **Usability:**The user interface should be intuitive with tooltips, help sections, and graphical representations (e.g., charts, word clouds).Cognitive results must be clearly labeled and interpreted for non-technical users.
- **Portability:**The system must be easily deployable on different environments using Docker containers. All components should support Linux, Windows, and Mac systems for development.
- **Maintainability:** The codebase should follow modular architecture to allow independent updates to the UI, backend, or ML models.Inline documentation and README files must be maintained with every component.
- **Reusability:**ML model modules and data processing pipelines should be structured as reusable Python classes or functions.
- **Testability:**Unit tests must be written for at least **80%** of backend functionality.GUI and integration tests should be conducted using tools like Selenium or PyTest.
- **Scalability:**The system must support horizontal scaling using container orchestration (e.g., Kubernetes) for handling high-load scenarios.
- **Interoperability:**The system should be able to ingest data from various formats (JSON, CSV, API) and integrate with external services for visualization or storage.

3.7.5 Business Rules

These rules govern the logical and procedural flow of how the system is used based on roles and responsibilities.

- **Role-Based Access Control (RBAC):**
 - **Admin:**Full control over system settings, model updates, and logs.
 - **Analyst:**upload data, run analyses, generate reports, and view logs.

- **Viewer:** Can view processed results and reports only.
- **Usage Policies:**
 - No user may upload or process **illegal, offensive, or hate-driven content**. Automated keyword filters will flag such data.
 - Analysts must agree to a **data ethics disclaimer** before using the tool.
- **Data Retention:**
 - Uploaded datasets will be retained for a maximum of **30 days** unless explicitly saved by the user.
 - Temporary files and cache must be cleared periodically to avoid storage overuse.
- **Analysis Quotas:**
 - Free-tier users (if applicable in future) will have a limit of 5 analyses/day to conserve server resources.

3.8. Preliminary Schedule and Budget

This section outlines the estimated timeline and budget required for the development, testing, and deployment of the project “**A Cognitive Psychology Approach to Analyse Information in Social Network Using Machine Learning.**” It is based on a standard academic mini-project lifecycle with appropriate time allocations for each phase.

3.8.1 Preliminary Schedule: the following table shows the schedule Table 3.3

Table 3.3 Schedule

Phase	Duration	Timeline	Milestones
Requirement Analysis	1 week	Week 1	Finalize SRS, use c flowchart
Research and Data Collection	1 week	Week 2	Literature survey, sele of cognitive psycho traits
Model Design and Training	2 weeks	Weeks 3–4	Develop ML models, training cycles
Backend and Data Processing Module	1 week	Week 5	Develop preproce pipeline, sentiment/cognitive par

Phase	Duration	Timeline	Milestones
Frontend Development	1 week	Week 6	Create UI components, input/output interfaces
Integration and Testing	1 week	Week 7	Combine modules, perform unit and integration tests
Final Review and Documentation	1 week	Week 8	Finalize report, create demo, run final tests

Total Project Duration: 8 weeks

3.8.2 Preliminary Budget: the following table shown the project Budget Table 3.4

Table 3.4 Budget

Component	Estimated Cost (INR)	Details
Development Tools (IDEs, Libraries)	₹0 (Open Source)	PyTorch, TensorFlow, Scikit-learn, Streamlit/Flask
Cloud/Server Costs (if hosted)	₹1,500	Cloud VM for training/inference (8-core CPU, 16GB RAM, 1 month)
Research Papers and Journals	₹500	Optional paid access to premium papers
Miscellaneous (printing, storage)	₹300	Report printing, flash drive, backups
Total Estimated Cost	₹2,300	

3.9 Appendices

This section provides additional reference information that supports and supplements the main Software Requirements Specification .

Following are the Glossary contents in Table 3.5

Appendix A: Table 3.5 Glossary

Term	Definition
SRS	Software Requirements Specification
ML	Machine Learning – a branch of AI focused on building systems that learn
Cognitive Psychology	A field of psychology concerned with mental processes like perception, memory, and reasoning
Sentiment Analysis	The use of NLP and ML to determine emotional sentiment behind words
Social Network Data	Data extracted from platforms like Twitter, Reddit containing user text
Streamlit	An open-source Python framework for creating apps for data science
RBAC	Role-Based Access Control – system of assigning permissions based on user roles
TBD	To Be Determined – placeholder for content that will be decided later

Appendix B: Analysis Models

The following diagrams and models support the analysis and design of the system:

- **Flowchart:** Depicts the logical process flow from data ingestion to ML analysis and user output.
- **Class Diagram:** Shows relationships between modules like DataProcessor, CognitiveModel, UserInterface (TBD).
- **Entity-Relationship Diagram (ERD)**
 - Entities: User, Dataset, AnalysisResult
 - Relationships: User uploads Dataset, which generates AnalysisResult.
- **State Transition Diagram (TBD):** Optional if system includes state-dependent processing (e.g., user roles).

Appendix C: To Be Determined List (TBD)

Following are the TBD contents in Table 3.6

Table 3.6 TBD

ID	Item Description	Expected Resolution Date
TBD-1	Choice between Flask or Stre for frontend	During Week 5
TBD-2	Final set of psychological tra be analyzed	End of Week 2
TBD-3	Final dataset for training testing	During Week 2
TBD-4	Whether cloud infrastructure be needed	After Week 3
TBD-5	Audit logging format compliance details	During Week 6

Signature of the Guide